Rahul Tumula

816-896-8105 | rahulthumula2077@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Software engineer with experience building backend services, cloud applications, and AI-powered systems across AWS and Azure. Skilled in API development, distributed workflows, document processing, and production reliability using Java, Python, and Ruby. Delivered AI pipelines processing 250K+ invoices, reduced API latency from 200 ms to 25 ms, and automated deployment workflows. Recent work focuses on user and agent authentication, biometric passkeys, and OAuth integrations at AWS.

PROFESSIONAL EXPERIENCE

Software Development Engineer II, Amazon Web Services - Herndon, VA (Aug 2025 - Present)

- Owned biometric passkey registration development, building Java Lambda APIs, DynamoDB challenge storage, MySQL migrations, and cross-region attestation integration for hardware-backed authentication.
- Implemented chained token delegation for local agents in a Ruby on Rails OAuth portal, adding session-security claims and authentication-method metadata; built Python integration tests for multi-hop and failure paths.
- Delivered self-service OAuth client registration, replacing hard-coded application configuration with managed client settings and validation across service APIs and the authentication portal.
- Expanded a password and account-unlock service into three isolated AWS partitions by unifying deployment pipelines and adding regional endpoints, certificate support, and integration tests.
- Migrated account-unlock SMS delivery to AWS End User Messaging with sender registration, phone-number validation, feature flags, and availability alarms; rolled out to the UK, Ireland, and Saudi Arabia.
- Remediated SSH certificate principal injection and reflected XSS through layered input validation and output encoding; added HTML sanitization for custom identity-provider email templates.
- Closed HTTP 5xx monitoring gaps with CloudWatch log-based alarms and added browser, OS, and device error-rate dashboards to improve authentication-service troubleshooting.

AI Engineer, Recipe-Costing - Miami, FL (Jul 2024 - Jul 2025)

- Built an Azure Vision and GPT-4o pipeline to extract data from **250K+ invoices**, achieving **92% F1** and reducing manual entry by **85%**.
- Orchestrated **10K items/day** with Azure Durable Functions, parallel ingestion, exponential backoff, and compensation patterns, achieving a failure rate below **0.01%**.
- Integrated GenAI services through FastAPI and .NET; built semantic search using Azure Cognitive Search and custom embeddings.

Volunteer AI Engineer, Healthi AI - Cambridge, MA (Jun 2024)

- Deployed Dockerized FastAPI inference endpoints on AWS with **sub-300 ms P95** latency and support for **100 concurrent requests**.

Graduate Teaching Assistant, University of Missouri-Kansas City (2023 - 2024)

- Guided students in statistical learning, model selection, and evaluation; graded assignments and projects.

Backend Developer, Tata Consultancy Services - Hyderabad, India (Jun 2021 - Dec 2022)

- Reduced Flask endpoint P95 latency from **200 ms to 25 ms** by introducing Redis caching for session and lookup data.
- Cut deployment time by **50%** with Azure DevOps CI/CD pipelines and automated pytest and Postman testing.

Backend Developer, Groom - Hyderabad, India (May 2020 - Jun 2021)

- Built a spaCy pipeline to summarize and categorize **100K+ insurance documents** with **87%** multi-label classification accuracy.
- Containerized PySpark ETL jobs and scheduled them on EKS using Kubernetes CronJobs and dynamic executor scaling.

SKILLS

- **Languages & frameworks:** Java, Python, Ruby on Rails, TypeScript/JavaScript, Go, SQL
- **Backend & cloud:** AWS Lambda, DynamoDB, MySQL, CDK, CloudWatch, Azure, Docker, Kubernetes
- **Identity & APIs:** OAuth 2.0/OIDC, WebAuthn, passkeys, REST APIs, Smithy, FastAPI
- **AI & data:** Azure OpenAI, RAG, vector search, LangGraph, PySpark

EDUCATION

- **University of Missouri-Kansas City** - Master of Science, Computer Science (May 2024)

CERTIFICATIONS

- Microsoft Certified: Azure AI Fundamentals; Azure AI Engineer Associate - Link
- NVIDIA: Generative AI/LLM Certification - Link
- Oracle Certified: Generative AI - Link